

## VIENTIANE WATTAY INTL, Lao People's Democratic Republic

WMO: 489400

Lat: 17.988N

Lon: 102.563E

Elev: 564

StdP: 14.40

Time Zone: 7.00 (E07)

Period: 03-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | 57.8       | 60.8 | 47.2                          | 49.1 | 67.3 | 50.7 | 56.0 | 66.7 | 11.7                  | 73.0 | 9.6 | 72.4 | 2.8                   | 0    | 0.291 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 4             | 18.5                   | 99.4            | 79.2 | 97.1 | 78.8 | 95.1 | 78.5 | 82.5                | 92.7 | 81.7 | 91.1 | 81.0 | 89.8 | 3.6                  | 180  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 80.3                            | 160.7 | 86.3 | 79.2 | 154.8 | 85.1 | 79.0 | 153.5 | 84.8 | 47.0          | 92.9 | 46.1 | 91.6 | 45.4 | 90.3 | 93.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |     |  |  |
|-------------------|------|-----|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|-----|--|--|
|                   |      |     | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |     |  |  |
| 1%                | 2.5% | 5%  | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max |  |  |
| (a)               | (b)  | (c) | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p) |  |  |
| (4) 10.3          | 8.3  | 7.1 | N/A                        | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |  |  |
| (5)               |      |     | DB                         | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |  |  |
|                   |      |     | WB                         | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |  |  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 81.1   | 73.5  | 78.0  | 82.2  | 85.7  | 84.9  | 84.3  | 83.2  | 82.7  | 83.0  | 82.2  | 79.0  | 74.7  |
|                                                                        | DBStd       | 5.17   | 4.59  | 4.55  | 4.71  | 3.87  | 3.09  | 2.55  | 2.61  | 2.39  | 1.98  | 2.33  | 3.85  | 4.26  |
|                                                                        | HDD50       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                        | HDD65       | 7      | 5     | 0     | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 1     |
|                                                                        | CDD50       | 11361  | 729   | 784   | 998   | 1070  | 1081  | 1029  | 1030  | 1014  | 990   | 1000  | 869   | 767   |
|                                                                        | CDD65       | 5892   | 269   | 364   | 534   | 620   | 616   | 579   | 565   | 549   | 540   | 535   | 419   | 302   |
|                                                                        | CDH74       | 63034  | 1959  | 3458  | 6040  | 7903  | 7294  | 6785  | 6157  | 5807  | 5865  | 5682  | 3846  | 2238  |
|                                                                        | CDH80       | 26552  | 595   | 1534  | 2985  | 4211  | 3423  | 2894  | 2336  | 2067  | 2183  | 2205  | 1442  | 677   |
| (14) Wind                                                              | WSAvg       | 3.3    | 3.1   | 3.3   | 3.3   | 3.5   | 3.4   | 3.3   | 3.3   | 3.2   | 3.1   | 3.1   | 3.2   | 3.3   |
| (15) Precipitation                                                     | PrecAvg     | 63.1   | 0.4   | 0.7   | 1.7   | 3.7   | 8.8   | 9.9   | 10.9  | 13.2  | 11.1  | 3.9   | 0.6   | 0.2   |
|                                                                        | PrecMax     | 81.5   | 3.1   | 4.7   | 5.3   | 11.5  | 20.5  | 18.7  | 25.0  | 25.1  | 24.0  | 12.3  | 4.2   | 1.6   |
|                                                                        | PrecMin     | 51.6   | 0.0   | 0.0   | 0.0   | 0.0   | 2.0   | 2.6   | 5.2   | 6.3   | 4.8   | 0.0   | 0.0   | 0.0   |
|                                                                        | PrecStd     | 8.7    | 0.7   | 0.9   | 1.4   | 2.8   | 4.1   | 3.8   | 4.3   | 4.4   | 4.4   | 2.6   | 1.1   | 0.4   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 91.3  | 96.7  | 100.2 | 103.6 | 100.5 | 97.0  | 96.8  | 93.6  | 93.6  | 93.5  | 93.2  | 90.9  |
|                                                                        |             | MCWB   | 71.6  | 72.0  | 76.6  | 80.9  | 81.2  | 80.8  | 80.5  | 80.4  | 80.5  | 79.0  | 78.0  | 75.4  |
|                                                                        | 2%          | DB     | 88.1  | 93.8  | 97.3  | 100.6 | 97.7  | 94.8  | 93.4  | 91.8  | 91.8  | 91.7  | 91.2  | 88.1  |
|                                                                        |             | MCWB   | 72.1  | 73.6  | 76.6  | 79.5  | 81.0  | 80.5  | 80.1  | 79.9  | 80.0  | 78.1  | 76.8  | 73.9  |
|                                                                        | 5%          | DB     | 86.0  | 91.6  | 95.2  | 98.3  | 95.2  | 92.9  | 91.5  | 90.3  | 91.0  | 90.4  | 89.5  | 86.1  |
|                                                                        |             | MCWB   | 71.6  | 73.5  | 76.6  | 79.4  | 80.5  | 80.2  | 79.6  | 79.5  | 79.7  | 77.4  | 75.7  | 72.7  |
|                                                                        | 10%         | DB     | 83.9  | 89.3  | 93.0  | 95.2  | 93.0  | 91.0  | 89.5  | 89.0  | 89.4  | 89.4  | 87.7  | 84.1  |
|                                                                        |             | MCWB   | 70.4  | 73.1  | 76.4  | 78.7  | 80.1  | 79.8  | 79.3  | 79.3  | 79.1  | 76.9  | 74.6  | 71.4  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 76.3  | 77.9  | 81.1  | 83.5  | 83.9  | 82.9  | 82.4  | 82.1  | 82.3  | 81.2  | 79.8  | 77.0  |
|                                                                        |             | MCDB   | 85.2  | 90.6  | 94.7  | 99.1  | 96.1  | 91.4  | 90.3  | 88.6  | 89.4  | 88.7  | 89.4  | 87.3  |
|                                                                        | 2%          | WB     | 74.4  | 76.4  | 79.7  | 81.8  | 82.4  | 81.8  | 81.1  | 81.0  | 81.2  | 80.0  | 78.4  | 75.4  |
|                                                                        |             | MCDB   | 83.5  | 88.8  | 93.0  | 96.5  | 93.8  | 89.9  | 88.6  | 88.0  | 88.3  | 87.4  | 87.2  | 84.7  |
|                                                                        | 5%          | WB     | 72.9  | 75.4  | 78.3  | 80.6  | 81.3  | 81.0  | 80.5  | 80.4  | 80.6  | 79.2  | 77.5  | 74.3  |
|                                                                        |             | MCDB   | 82.1  | 87.5  | 91.1  | 94.1  | 91.9  | 88.9  | 87.8  | 87.3  | 87.5  | 86.2  | 85.5  | 83.1  |
|                                                                        | 10%         | WB     | 71.8  | 74.3  | 77.2  | 79.6  | 80.4  | 80.4  | 79.8  | 79.8  | 79.9  | 78.5  | 76.5  | 72.9  |
|                                                                        |             | MCDB   | 80.2  | 85.6  | 89.4  | 91.8  | 90.1  | 88.0  | 86.5  | 86.1  | 86.3  | 85.2  | 84.0  | 81.3  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 19.3  | 20.7  | 19.6  | 18.5  | 15.6  | 12.9  | 12.0  | 11.6  | 12.7  | 14.3  | 17.1  | 18.5  |
|                                                                        |             | MCDBR  | 21.7  | 24.0  | 23.0  | 22.0  | 19.1  | 15.6  | 15.3  | 14.4  | 15.1  | 16.0  | 18.2  | 19.9  |
|                                                                        |             | MCWBR  | 8.8   | 8.4   | 8.0   | 7.6   | 7.0   | 5.4   | 5.1   | 5.3   | 5.5   | 5.4   | 6.8   | 8.0   |
|                                                                        | 5% WB       | MCDBR  | 19.2  | 21.1  | 20.9  | 20.3  | 17.2  | 13.9  | 13.1  | 12.8  | 13.5  | 14.8  | 16.3  | 18.7  |
|                                                                        |             | MCWBR  | 8.4   | 8.4   | 8.4   | 7.6   | 7.0   | 5.5   | 5.1   | 5.4   | 5.6   | 5.5   | 6.8   | 8.2   |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.465 | 0.575 | 0.701 | 0.666 | 0.504 | 0.456 | 0.445 | 0.464 | 0.507 | 0.545 | 0.454 | 0.459 |
|                                                                        | taud        |        | 2.118 | 1.841 | 1.602 | 1.717 | 2.169 | 2.331 | 2.358 | 2.286 | 2.151 | 2.001 | 2.232 | 2.154 |
|                                                                        | Ebn at Noon |        | 258   | 234   | 211   | 220   | 256   | 266   | 269   | 266   | 255   | 239   | 260   | 255   |
|                                                                        | Edh at Noon |        | 47    | 66    | 86    | 77    | 48    | 41    | 40    | 43    | 49    | 55    | 42    | 44    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 1459  | 1603  | 1632  | 1780  | 1665  | 1525  | 1438  | 1419  | 1465  | 1525  | 1522  | 1413  |
|                                                                        | RadStd      |        | 101   | 90    | 96    | 92    | 123   | 120   | 99    | 101   | 80    | 97    | 62    | 64    |

## Historical Trends

|                              | DBAvg | Heating |     |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|-----|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only            | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (10 neighbors) | +0.49 | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page